

EXCEL DATA ANALYTICS – FINAL PROJECT GUIDELINE

1. Project Title

Practical Excel Data Analytics Project

Course: Excel for Data Analysis

Project Type: Individual Project

Students: All assigned students

Software: Microsoft Excel

2. Project Objective

The purpose of this project is to assess the students' ability to use Microsoft Excel to:

Collect → Prepare → Clean → Analyze → Visualize → Present Data

Students must apply the Excel skills learned throughout the course to a real-world dataset.

The course emphasizes that Excel is not only for entering data, but also for organizing data, analyzing it, and supporting informed decisions.

3. Individual Project Requirement

Every student must complete a different project.

- Each student will receive **one project title** from the lecturer.
- Students **must not use the same project title/dataset** unless approved by the lecturer.
- **Important**

5. Project Sections

Every student must complete the following sections.

Part A — Dataset Introduction

Students must explain:

1. Project title
2. Dataset name
3. Source of dataset
4. Purpose of the dataset
5. Number of records
6. Number of variables
7. Description of important variables

Part B — Data Preparation

Students must prepare the original dataset before analysis.

They should demonstrate:

✓ **import**

✓ **Data types**

Students should identify:

- Text
- Number
- Date
- Currency
- Percentage

The course specifically covers these Excel data types and number formats.

✓ **Formatting**

Students should apply appropriately:

- Number formatting
- Currency
- Percentage
- Date
- Table formatting
- Column widths
- Headings

Part B — Data Cleaning

This section is **mandatory**.

Students must identify and handle data-quality problems.

7.1 Duplicate Records

Students should:

1. Identify duplicates
2. Investigate them
3. Remove confirmed duplicates

The course explains that duplicates can affect totals, averages, PivotTables, charts, and dashboards.

7.2 Missing Values

Students should identify:

- Blank cells
- N/A
- Unknown
- NULL
- -

They must explain how they handled the missing values.

The course provides several approaches, including deleting a record when appropriate, replacing with a default value, and replacing missing numeric values with an average.

Part C — Text Cleaning

Students must demonstrate at least **3 text functions** where appropriate.

Part D — Data Analysis Using Formulas

Every student must demonstrate appropriate Excel formulas.

Basic calculations

At minimum:

- SUM
- AVERAGE
- COUNT
- MIN
- MAX

10. Conditional Functions

Students must use at least **two** of:

- SUMIF
- COUNTIF
- AVERAGEIF

The course explicitly introduces these three conditional functions.

11. Logical Functions

Students must use at least **one** logical analysis.

Possible functions:

- IF
- IFS
- AND
- OR

The course uses student results as a practical application of IF and grading with nested IF/IFS.

Part E — Date Analysis

If the dataset contains dates, students should demonstrate appropriate date functions.

Possible functions:

- TODAY()
- NOW()
- DAY()
- MONTH()
- YEAR()
- DATE()

The course also introduces working-day calculations with NETWORKDAYS.

Part E — Sorting and Filtering

Students must demonstrate:

Sorting

- A–Z
- Z–A
- Smallest–Largest
- Largest–Smallest
- Oldest–Newest
- Newest–Oldest

Filtering

Students should answer questions using filters.

The course covers both basic and multilevel sorting and explains that filtering displays only records matching selected conditions.

Part F — Conditional Formatting

Students must use at least **two** conditional-formatting techniques.

The course specifically demonstrates these techniques for student performance data.

Part G — PivotTables

This is a **mandatory section**.

Each student must create **PivotTables** based on projects:

The course teaches PivotTables as a method for quickly summarizing and analyzing large datasets.

Students must understand:

- Rows
- Columns
- Values
- Filters

These are the four main PivotTable areas taught in the course.

Part H — Data Visualization

Each student must create at least: **5 Charts**

Students should select charts according to the question they want to answer.

Important Rule

Students should normally create charts from **PivotTables**, rather than selecting hundreds of raw records unnecessarily.

Part I — KPI Cards

Each student must create **4 KPI cards**. The KPI must be relevant to the project.

Part J — Interactive Slicers

Each student must create Slicers

The slicers should be based on the dataset.

Slicers should interact with the PivotTables/PivotCharts.

Part K — Final Dashboard

Every student must create **Two professional Excel dashboard.**

Minimum dashboard requirements

4 KPI Cards

4 Charts

2–3 Slicers

Recommended structure:

PROJECT DASHBOARD			
Student Name			
KPI 1	KPI 2	KPI 3	KPI 4
CHART 1			
CHART 2		CHART 3	
Filters: [Category] [City] [Status]			

Part M — Findings and Recommendations

This is very important. Students should not only create charts.

They must **interpret the data**.

Each student must provide:

At least 5 findings

Example:

1. Electronics generated the highest revenue.
2. Mogadishu had the highest number of transactions.
3. Sales increased during the third quarter.
4. Product A generated the highest profit.
5. Returning customers contributed more revenue than new customers.

Then provide:

3 recommendations

Example:

1. Increase stock for high-performing products.
2. Focus marketing activities on high-performing cities.
3. Develop strategies to increase repeat customers.

Required Excel Worksheets

I recommend that **every student uses the same workbook structure**:

- Sheet 1 → Raw_Data = **Do not modify the original data** ----- Raw_Data
- Sheet 2 → Clean_Data = Cleaned and standardized dataset
- Sheet 3 → Analysis = Formulas and calculations
- Sheet 4 → PivotTables = All PivotTables.
- Sheet 5 → Dashboard = Final visual presentation
- Sheet 6 → Findings = Written interpretation and recommendations.

23. Final Submission

Each student must submit:

1. Excel Workbook .xlsx

Containing all required worksheets.

2. Short Project Report = Recommended: 5–8 pages

3. Dashboard Screenshot = Include the final dashboard in the report.

4. Presentation

Each student should briefly explain:

- Dataset
- Cleaning
- Formulas
- PivotTables
- Charts
- KPIs
- Slicers
- Findings
- Recommendations

Lecturer's key rule

your course is **Excel for Data Analysis**, not simply Excel dashboard design. The course material emphasizes preparing and cleaning data, including duplicates and missing values, before analysis.

The overall project should therefore demonstrate:

**RAW DATA → CLEAN DATA → FORMULAS → PIVOTTABLES → CHARTS → KPIs
→ SLICERS → DASHBOARD → FINDINGS**